

Julius Liu

Software Developer | TypeScript, React, AI Agents

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WORK EXPERIENCE

Software Developer Co-op, Nokia — Ottawa ON

Sept 2025 – Present

- Delivered end-to-end functionality across a distributed system, designing backend services and APIs with comprehensive unit and E2E test coverage.
- Built a WebSocket-based SSH console for Kubernetes containers, enabling real-time terminal access and SFTP through a bastion architecture, with concurrent stream handling for reliable full-duplex communication.
- Led improvements to frontend accessibility and testability by introducing ARIA labels and semantic attributes, enabling Playwright-based E2E testing and improving selector reliability.
- Implemented online and offline evaluation pipelines with AI agents using Langfuse, integrating dataset-driven testing and CI workflows to evaluate and improve LLM-based agent performance.

EDUCATION

Computer Programming Diploma, Algonquin College — Ottawa ON

Aug 2026 (Expected)

- Cumulative GPA: 3.99/4.0, Dean's Honour List

SKILLS

- **Backend:** Python (FastAPI), Node.js, Express.js, Hono, REST APIs
- **Frontend:** React, Next.js, TypeScript, JavaScript, Vite, Webpack, ESLint, npm
- **Testing:** Vitest, Playwright, Postman, Bruno
- **AI & Agents:** Langfuse, LLM evaluation, MCP, Google ADK, GitHub Copilot, Claude Code
- **Tools & Platforms:** Docker, Kubernetes, Git, CI/CD, Linux, GCP

PROJECTS

vscode-containerlab (Open-Source VS Code Extension Contribution)

TypeScript, React

- Contributed a merged feature to an open-source VS Code extension for network lab topology management, adding a floating panel in the React-based topology viewer for configuring link impairment settings.
- Refactored the extension's webview messaging layer by extracting shared message schemas and separating extension commands into dedicated handler hooks, improving type safety and reducing coupling between the extension host and webview.

GitLab MR Review Agent

Python, Google ADK

- Developed a containerized AI agent integrating with a custom GitLab MCP server to retrieve diffs, metadata, and pipeline context for automated merge request review.
- Designed and orchestrated a modular multi-agent system to analyze code changes, documentation, and CI pipelines, generating structured and actionable review feedback.